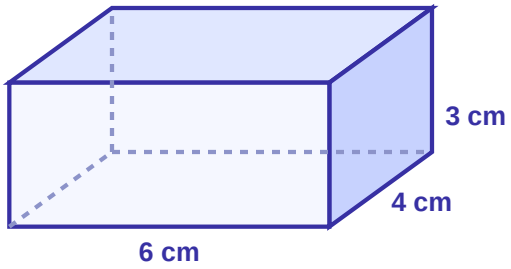
Name: _____ **Date:** _____

DIRECTIONS

This Independent Practice is for Lesson 4.17.03.
Write your answers on the lines. Show your work in the boxes.
A box has 12 edges: 4 lengths, 4 widths and 4 heights.
Total edge length = 4 × length + 4 × width + 4 × height.

1 A straw frame

Cora builds a straw frame for this box. Answer each part.



(a) How many straws are 6 cm long? _____

(b) How many straws are 4 cm long? _____

(c) How many straws are 3 cm long? _____

(d) How much straw does the whole frame use? _____ cm

2 Parallel or perpendicular?

The top face of this box is shaded. One edge is drawn extra thick.

(a) How many edges are parallel to the thick edge? (b) How many edges are perpendicular to it and touch it?

(c) How many faces are perpendicular to the shaded face?

Hint: perpendicular edges meet it at a square corner.

(a) _____ **(b)** _____ **(c)** _____

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3 Cube frames

(a) Hazel makes a cube frame from straws. Each straw is 5 cm long. How much straw does she use?

(b) Felix bends 72 cm of wire into a cube frame with nothing left over. How long is each edge?

SHOW YOUR WORK

(a) _____ cm **(b)** _____ cm

4 Find the mistake

A gift box is 9 cm long, 5 cm wide and 2 cm tall. Kai says the ribbon around every edge is 9 + 5 + 2 = 16 cm. Explain his mistake. Then find the right total.

KAI'S WORK: 9 + 5 + 2 = 16 cm

Total: _____ cm **Explain:** _____ _____

5 Challenge: work backward

A box frame uses 48 cm of straw. The box is 5 cm long and 4 cm wide. How tall is it?

THINK: HOW MUCH STRAW DO THE LENGTHS AND WIDTHS USE?

Height: _____ cm

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